

Dimitrios Papageorgiou

List of Publications

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Journal Articles

- [1] Pietro A. Refosco, Christopher Edwards, and **Dimitrios Papageorgiou**, "Conditions for boundedness of under-tuned super-twisting sliding mode control loops", *IEEE Transactions on Automatic Control*, vol. PP, no. 99, 2026. DOI: 10.1109/TAC.2026.3663109
- [2] Alessio Dallabona, Patrik Schermann, Mogens Blanke, and **Dimitrios Papageorgiou**, "Dynamic Input Mapping Inversion to Eliminate Algebraic Loops in Hydraulic Actuator Control", *IEEE Transactions on Control Systems Technology*, vol. 34, no. 2, pp. 979–989, 2026. DOI: 10.1109/TCST.2025.3641638
- [3] Adrià Mompou Alepuz, **Dimitrios Papageorgiou**, and Silvia Tolu, "Online learning and control of nonlinear delayed systems with efficient signal history encoding", *Physica Scripta*, vol. 100, 2025. DOI: 10.1088/1402-4896/ae2166
- [4] Alessio Dallabona, Mogens Blanke, Henrik C. Pedersen, and **Dimitrios Papageorgiou**, "Fault diagnosis and prognosis capabilities for wind turbine hydraulic pitch systems", *Mechanical Systems and Signal Processing*, vol. 224, 2025. DOI: 10.1016/j.ymssp.2024.111941
- [5] Morgan Lou'edec, Sol'ene Boudot, **Dimitrios Papageorgiou**, and Roberto Galeazzi, "Increasing the resilience of MASS remote pilot's situational awareness using land-based sensors", *Journal of Physics: Conference Series*, vol. 3123, no. 1, 2025. DOI: 10.1088/1742-6596/3123/1/012065
- [6] Namireddy Praveen Reddy, Roger Skjetne, Oliver Stugard Os, and **Dimitrios Papageorgiou**, "A Comparison of the State-of-the-Art Reinforcement Learning Algorithms for Health-Aware Energy & Emissions Management in Zero-emission Ships", *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, vol. 5, no. 1, pp. 149 – 166, 2024. DOI: 10.1109/JESTIE.2023.3331230
- [7] Adrià Mompou Alepuz, **Dimitrios Papageorgiou**, and Silvia Tolu, "Brain-inspired biomimetic robot control: a review", *Frontiers in Neurorobotics*, vol. 18, 2024. DOI: 10.3389/fnbot.2024.1395617
- [8] **Dimitrios Papageorgiou**, Guðrún Þóra Sigurðardóttir, Egidio Falotico, and Silvia Tolu, "Sliding-mode control of a soft robot based on data-driven sparse identification", *Control Engineering Practice*, vol. 144, 2024. DOI: 10.1016/j.conengprac.2023.105836
- [9] Mogens Blanke, Nicholas Hansen, Kjeld Dittmann, Thomas Thuesen Enevoldsen, Dimitrios Dagdilelis, Frederik Emil Thorsson Saabye Sch'oller, Martin Krarup Plenge-Feidenhans'l, Jonathan Binner Beckett, **Dimitrios Papageorgiou**, and Roberto Galeazzi, "GreenHopper: The Danish spearhead towards autonomous waterborne mobility", *Journal of Physics: Conference Series*, vol. 2867, 2024. DOI: 10.1088/1742-6596/2867/1/012035
- [10] **Dimitrios Papageorgiou** and Christopher Edwards, "On the behaviour of under-tuned super-twisting sliding mode control loops", *Automatica*, vol. 135, 2022. DOI: 10.1016/j.automatica.2021.109983
- [11] Rasmus Esbjørn Nielsen, **Dimitrios Papageorgiou**, Lazaros Nalpantidis, Bugge T. Jensen, and Mogens Blanke, "Machine Learning Enhancement of Manoeuvring Prediction for Ship Digital Twin using Full-scale Recordings", *Ocean Engineering*, vol. 257, 2022. DOI: 10.1016/j.oceaneng.2022.111579
- [12] **Dimitrios Papageorgiou**, Nicholas Hansen, Kjeld Dittmann, and Mogens Blanke, "Anticipation of ship behaviours in multi-vessel scenarios", *Ocean Engineering*, vol. 266, no. Part 2, 2022. DOI: 10.1016/j.oceaneng.2022.112777
- [13] **Dimitrios Papageorgiou**, Mogens Blanke, Hans Henrik Niemann, and Jan H. Richter, "Online friction parameter estimation for machine tools", *Advanced Control for Applications: Engineering and Industrial Systems*, 2020. DOI: 10.1002/adc2.28

- [14] **Dimitrios Papageorgiou**, Mogens Blanke, Hans Henrik Niemann, and Jan H. Richter, "Robust Backlash Estimation for Industrial Drive-Train Systems\textemdashTheory and Validation", *IEEE Transactions on Control Systems Technology*, vol. 27, no. 5, pp. 1847–1861, 2019. DOI: 10.1109/TCST.2018.2837642
- [15] **Dimitrios Papageorgiou**, Mogens Blanke, Hans Henrik Niemann, and Jan H. Richter, "Adaptive and Sliding Mode Friction-Resilient Machine Tool Positioning - Cascaded Control Revisited", *Mechanical Systems and Signal Processing*, vol. 132, pp. 35–54, 2019. DOI: 10.1016/j.ymssp.2019.06.015
- [16] **Dimitrios Papageorgiou**, Mogens Blanke, Hans Henrik Niemann, and Jan H. Richter, "Friction-resilient position control for machine tools\textemdashAdaptive and sliding-mode methods compared", *Control Engineering Practice*, vol. 75, pp. 69–85, 2018. DOI: 10.1016/j.conengprac.2018.03.017

Conference Papers

- [1] Antoine Thibault Vi\`e, Leonid Fridman, Roberto Galeazzi, and **Dimitrios Papageorgiou**, "Multi-layer barrier adaptation of the discrete-time super-twisting controller", *Proceedings of IEEE 18th International Workshop on Variable Structure Systems*, pp. 143–48, 2026. DOI: 10.1109/VSS69650.2026.11655757
- [2] Antoine Thibault Vi\`e, Leonid Fridman, Roberto Galeazzi, and **Dimitrios Papageorgiou**, "Multi-layer barrier function-based adaptive super-twisting controller", *Proceedings of IEEE 64th Conference on Decision and Control*, pp. 7932–7938, 2026. DOI: 10.1109/CDC57313.2025.11312295
- [3] Jeppe H. Mikkelsen, Thomas T. Enevoldsen, Bugge T. Jensen, Michael Jeppesen, Roberto Galeazzi, and **Dimitrios Papageorgiou**, "Closed Form Modelling and Identification of Banking Effects in Confined Waters", *Proceedings of 2026 European Control Conference*, pp. 3277–82, 2026.
- [4] Yaqub A. Prabowo, Peter N. Hansen, **Dimitrios Papageorgiou**, and Roberto Galeazzi, "Quantifying Good Seamanship: A Framework for Risk-Aware Trajectory Planning in Autonomous Vessel Navigation", 2025.
- [5] Yaqub A. Prabowo, Peter N. Hansen, Jens B. Jensen, **Dimitrios Papageorgiou**, and Roberto Galeazzi, "Good Seamanship Online Trajectory Planner for Autonomous Ship", *IFAC-PapersOnLine*, vol. 59, no. 22, pp. 359–364, 2025. DOI: 10.1016/j.ifacol.2025.11.659
- [6] Alessio Dallabona, Mogens Blanke, and **Dimitrios Papageorgiou**, "Sensorless Friction Estimation for Condition Monitoring of Wind Turbine Hydraulic Pitch System", *Proceedings of 6th International Conference on Control and Fault-Tolerant Systems*, pp. 95–100, 2025. DOI: 10.1109/SysTol66549.2025.11267321
- [7] Antoine Thibault Vi\`e, Roberto Galeazzi, and **Dimitrios Papageorgiou**, "Extended Time Varying Multi-Cluster Fluctuating Two-Ray Fading Model for Maritime Environment", *Elsevier*, vol. 59, no. 22, pp. 734–739, 2025. DOI: 10.1016/j.ifacol.2025.11.722
- [8] Adri\`a Momp\`o Alepuz, **Dimitrios Papageorgiou**, and Silvia Tolu, "Adaptive Learning-based Predictor for Enhanced Control of Assistive Soft Robots", *Proceedings of 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2025. DOI: 10.1109/EMBC58623.2025.11252603
- [9] Duc Trung Hoang, **Dimitrios Papageorgiou**, Peter N. Hansen, \Asmus S.\ Christiansen, and Roberto Galeazzi, "Predictive Maintenance of Road Networks via Spatial Clustering and Scheduling: A Data-Driven Approach", *Proceeding of 6th International Conference on Control and Fault-Tolerant Systems*, pp. 429–436, 2025. DOI: 10.1109/SYSTOL66549.2025.11267347
- [10] Alessio Dallabona, Mogens Blanke, and **Dimitrios Papageorgiou**, "Friction Estimation for Condition Monitoring of Wind Turbine Hydraulic Pitch System", *IFAC-PapersOnLine*, vol. 58, no. 4, pp. 598–603, 2024. DOI: 10.1016/j.ifacol.2024.07.284
- [11] Nicholas Hansen, Yaqub Aris Prabowo, **Dimitrios Papageorgiou**, and Roberto Galeazzi, "Uncertainty-Aware Decision Making for Safe Navigation of Autonomous Ships", *IFAC-PapersOnLine*, vol. 58, no. 20, pp. 223–228, 2024. DOI: 10.1016/j.ifacol.2024.10.058
- [12] Yaqub Aris Prabowo, Nicholas Hansen, **Dimitrios Papageorgiou**, and Roberto Galeazzi, "Good Seamanship Score Quantification in Complex and Congested Waterways.", *IFAC-PapersOnLine*, vol. 58, no. 20, pp. 341–346, 2024. DOI: 10.1016/j.ifacol.2024.10.077

- [13] Alessio Dallabona, Mogens Blanke, and **Dimitrios Papageorgiou**, "Model-free sliding mode control for hydraulic actuators under high load", *Proceedings of 17th International Workshop on Variable Structure Systems*, pp. 314–320, 2024. DOI: 10.1109/VSS61690.2024.10753380
- [14] Fatemeh Akbarian, **Dimitrios Papageorgiou**, Seyedezahra Chamideh, Jeppe Heini Mikkelsen, Peter Iwer Hoedt Karstensen, and Maria Kihl, "Enhancing Autonomous Vehicles System Security: Advanced Attack Detection for Robust Safeguarding", *Proceedings of 10th International Conference on Control, Decision and Information Technologies*, pp. 730–737, 2024. DOI: 10.1109/CoDIT62066.2024.10708181
- [15] **Dimitrios Papageorgiou** and Ilmar Santos, "Sliding mode control of active magnetic bearings - a cascaded architecture", *Elsevier*, vol. 56, no. 2, pp. 2480–2485, 2023. DOI: 10.1016/j.ifacol.2023.10.1227
- [16] Namireddy Praveen Reddy, Roger Skjetne, and **Dimitrios Papageorgiou**, "A Decentralized Droop-Based Power Management System for Ship Power Systems Using Hybrid Dynamical Systems Framework", *Proceedings of the 2nd International Conference on Ocean, Offshore and Arctic Engineering*, vol. 5, 2023. DOI: 10.1115/OMAE2023-102570
- [17] Seyed Mohammad Asadzadeh, **Dimitrios Papageorgiou**, and Nils Axel Andersen, "Nonlinear control of a selective catalytic reduction unit for a bio-fueled cogeneration plant", *IFAC-PapersOnLine*, vol. 56, no. 2, pp. 3923–3929, 2023. DOI: 10.1016/j.ifacol.2023.10.1330
- [18] Namireddy Praveen Reddy, Yuxuan Cai, Roger Skjetne, and **Dimitrios Papageorgiou**, "Voltage Sensor Fault Detection in Li-ion Battery Energy Storage Systems", *Proceedings of 2022 IEEE Transportation Electrification Conference and Expo*, pp. 1314–1320, 2022. DOI: 10.1109/ITEC53557.2022.9813853
- [19] Peter Nicholas Hansen, Thomas T. Enevoldsen, **Dimitrios Papageorgiou**, and Mogens Blanke, "Autonomous Navigation in Confined Waters - A COLREGs Rule 9 Compliant Framework", *Elsevier*, vol. 55, no. 31, pp. 222–228, 2022. DOI: 10.1016/j.ifacol.2022.10.435
- [20] **Dimitrios Papageorgiou**, "Towards Prescribed Accuracy in Under-tuned Super-Twisting Sliding Mode Control Loops - Experimental Verification", *Proceedings of 2022 American Control Conference*, pp. 2503–2508, 2022. DOI: 10.23919/ACC53348.2022.9867161
- [21] Kjeld Dittmann, Nicholas Hansen, **Dimitrios Papageorgiou**, and Mogens Blanke, "Autonomy for Ships: A Sovereign Agents Architecture for Reliability and Safety by Design", *Proceedings of 5th International Conference on Control and Fault-Tolerant Systems*, pp. 50–57, 2021. DOI: 10.1109/SysTol52990.2021.9595889
- [22] Kjeld Dittmann, Nicholas Hansen, **Dimitrios Papageorgiou**, Signe Jensen, Marie L^utzen, and Mogens Blanke, "Autonomous Surface Vessel with Remote Human on the Loop: System Design for STCW Compliance", *IFAC-PapersOnLine*, vol. 54, no. 16, pp. 224–231, 2021. DOI: 10.1016/j.ifacol.2021.10.097
- [23] Zoltan Mark Pinter, **Dimitrios Papageorgiou**, Gunnar Rohde, Mattia Marinelli, and Chresten Tr^{ae}holt, "Review of Control Algorithms for Reconfigurable Battery Systems with an Industrial Example", *Proceedings of 56th International Universities Power Engineering Conference*, 2021. DOI: 10.1109/UPEC50034.2021.9548259
- [24] Fabio Spagnolo, **Dimitrios Papageorgiou**, Roberto Galeazzi, Jesper Sandberg Thomsen, and Kim Hylling S^lorensen, "Extremum Seeking Control for Multi-Rotor Wind Turbine in the Full-Load Region.", *Elsevier*, vol. 53, no. 2, pp. 5386–5391, 2021. DOI: 10.1016/j.ifacol.2020.12.1525
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- [26] Peter Nicholas Hansen, **Dimitrios Papageorgiou**, Mogens Blanke, Roberto Galeazzi, M. L^utzen, J. Mogens, M. Bennedsen, and D. Hansen, "COLREGs-based Situation Awareness for Marine Vessels - a Discrete Event Systems Approach", *Elsevier*, vol. 53, no. 2, pp. 14501–14508, 2021. DOI: 10.1016/j.ifacol.2020.12.1453
- [27] **Dimitrios Papageorgiou**, Mogens Blanke, Marie L^utzen, Mette Bennedsen, John Mogens, and S^lorensen, "Parallel Automaton Representation of Marine Crafts\textquoteright COLREGs-based Manoeuvring Behaviours", *IFAC-PapersOnLine*, vol. 52, no. 21, pp. 103–110, 2019. DOI: 10.1016/j.ifacol.2019.12.291

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Theses

[1] **Dimitrios Papageorgiou**, “High Accuracy Nonlinear Control and Estimation for Machine Tool Systems”, *Technical University of Denmark, Department of Electrical Engineering*, 2017.